

Project Initialization and Planning Phase

Date	16 June 2025
Team ID	SWTID1749707061
Project Title	Online Payments Fraud Detection Using Machine Learning
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To develop a machine learning-based system capable of detecting fraudulent online payment transactions in real-time with high accuracy.
Scope	The project involves data collection, preprocessing, machine learning model development, evaluation, and deployment of a fraud detection system with a user interface. It excludes legal enforcement or integration with third-party banking APIs.
Problem Statement	
Description	Online payment fraud is a major concern in digital financial services. Traditional systems fail to identify complex or new fraudulent behavior, putting user trust and safety at risk.
Impact	Implementing a real-time fraud detection system using ML can help prevent financial losses, increase trust in digital transactions, and improve operational efficiency for service providers.
Proposed Solution	
Approach	Use supervised machine learning techniques (e.g., Random Forest, XGBoost) to analyze transaction patterns. The pipeline includes dataset collection, preprocessing, model training, evaluation, and deploying a web interface for prediction.

Key Features	- Real-time fraud detection - Interactive web interface for predictions - Data visualization for fraud analysis - High precision and recall using optimized models
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU for model training and testing	Intel i7 / 2 x NVIDIA V100 GPUs
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Backend and ML frameworks	Flask
Libraries	Additional libraries for data handling & ML	Pandas, NumPy, Matplotlib, XGBoost
Development Environment	Tools used for development and version control	Jupyter Notebook, VS Code, Git
Data		
Data	Source, size, format	Kaggle dataset: "Online Payment Fraud Detection", CSV format, ~600,000 records